CYBERCRIME INVESTIGATION & FORENSIC II

CYS340

DIGITAL FORENSICS | OVERVIEW

LECTURE 1

INTRODUCTION

- **Cybercrime** is any illegal or criminal activity carried out by means of computers or the internet.

CYBERCRIME INVESTIGATORS

- **Cybercrime investigators** are law enforcement professionals who specialize in investigating computer-related crimes.
- According to Act no. 2 of 2021 PART I (Preliminary Provision) "**Law Enforcement**" means –

a) a police officer above the rank of sub-inspector;

b) an officer of the Anti-Corruption Commission;

c) an officer of the Drug Enforcement Commission;

d) an officer of the Zambia Security Intelligence Service; and

e) any other person appointed by the Minister for purposes of this Act;

CYBERCRIME INVESTIGATORS (CONTINUES…)

- They use their knowledge of **computer science, law,** and **investigative techniques** to gather evidence and identify suspects.
- Cybercrime investigators may work for **law enforcement agencies, government agencies,** or **private companies.**
- Their job duties may include conducting **investigations, preparing reports,** and **providing testimony in court.**

CYBERCRIME INVESTIGATION

- Is the process of using digital forensics to identify and prosecute cybercriminals, or
- Is the process of identifying, analyzing, and mitigating computer-based crimes and other forms of malicious activity that occur in cyberspace.
- It involves the use of specialized tools and techniques to investigate various types of cybercrimes, such as **hacking**, **phishing**, **malware**, **data breaches**, and **identity theft**.

TOP 5 CYBERCRIMES

1. **Phishing and Scam**

- **Phishing** is a form of social engineering and a scam where attackers deceive people into revealing sensitive information or installing malware such as viruses, worms and trojans through attachments or links sent via email by impersonating a legitimate organization or entity.
- **Scam** is a dishonest scheme; a fraud, similar to swindle, trick, etc.

TOP 5 CYBERCRIMES (CONTINUES…)

2. **Identity Theft**

- The fraudulent practice of using another person's name and personal information in order to obtain credit, loans, etc.

3. **Ransomware Attack**

- Is a type of malware that encrypts the victim's personal data until a ransom is paid.

TOP 5 CYBERCRIMES (CONTINUES…)

4. **Hacking/Misusing Computer Networks**

- The gaining of unauthorized access to data in a system or computer.

5. **Internet Fraud**

- Also known as **online fraud**, refers to any deceptive practice that uses the internet to trick individuals or businesses into giving up personal information, money, or property, often by presenting false offers, scams, or misleading information through emails, websites, or online platforms to gain financial advantage.

PROTECTING AGAINST CYBERCRIMES

- It is important for businesses and individuals to take **proactive measures** to protect against these types of cybercrimes, such as **implementing strong security measures** and **staying vigilant against suspicious activity.**

1. Installing and regularly updating **anti-virus software**
2. Use **Multi-factor authentication (MFA)**
3. Use **strong passwords**

TYPES OF CYBER CRIMINALS

1. **Hackers /Attackers**

- Individuals who use their expertise to gain unauthorized access to computer systems or networks for various reasons such as stealing data or causing damage.
- They can be motivated by financial gain, political or social causes, or simply a desire to test their skills.

TYPES OF CYBER CRIMINALS (CONTINUES…)

2. **Insiders**

- Individuals who have authorized access to computer systems or networks but misuse their access for personal gain or to harm the organization.

3. **Organized Crime Groups**

- Groups of individuals who engage in cybercrime for financial gain or other criminal activities.

TYPES OF CYBER CRIMINALS (CONTINUES…)

4. **National / States**

- Governments or state-sponsored groups use cyber-attacks for political or military purposes.
- Nation-states can use cyber-attacks to gather intelligence, disrupt systems, or gain a strategic advantage over other countries.

TYPES OF CYBER CRIMINALS (CONTINUES…)

5. **Cyberterrorists**

- Individuals or groups that use cyber-attacks to cause fear, disrupt systems, or harm individuals or organizations.
- They can be motivated by political or social causes.

DIGITAL FORENSICS

- **Cybercrime investigation** techniques involve a combination of **technical** and **non-technical** methods for gathering evidence and identifying suspects.
- One of the most important techniques used in cybercrime investigation is **digital forensics**.
- **Digital forensics** involves the collection, preservation, and analysis of digital evidence.

DIGITAL FORENSICS (CONTINUES…)

- Digital forensics can include;

1. The **recovery of deleted files,**
2. **analyzing metadata**, and
3. **examining network traffic logs.**

CYBERCRIME INVESTIGATION TECHNIQUES

1. **Conducting interviews with witnesses**
2. Reviewing **surveillance footage**
3. **Analyzing financial records to track the flow of money**
4. Use **social engineering techniques** to gather information about suspects, such as posing as potential victims or using fake social media profiles to gain access to information.
5. **Collaboration and Specialization Unit** – Cybercrime investigations often involve multiple agencies and organizations, and it's important for investigators to work together to share information and resources.
6. **Digital Forensics**

CYBERCRIME INVESTIGATION TOOLS

DIGITAL FORENSIC SOFTWARE

1. EnCase
2. FTK
3. Autopsy

NETWORK ANALYSIS TOOLS

1. Wireshark
2. Tcpdump
3. Netscout

MALWARE ANALYSIS TOOLS

1. IDA Pro
2. OllyDbg
3. Binary Ninja

CYBERCRIME INVESTIGATION TOOLS (CONTINUES…)

PASSWORD RECOVERY TOOLS

1. Cain
2. Abel
3. John the Ripper
4. Hashcat

SOCIAL MEDIA ANALYSIS TOOLS

1. Hootsuite
2. Followerwonk
3. Mention

DIGITAL FORENSICS

LECTURE 2

INTRODUCTION

- **Forensics** is the scientific analysis of people, places and things to collect evidence during crime investigations, that helps to prove innocence or guilt in court, or
- Is the art and science of investigating digital crimes and breaches.
- **Computer forensic** (digital forensics or cyber forensics) is a branch of digital forensic science that uses technology and investigative techniques to analyze data from computer systems, networks, wireless communications, and storage devices.

CHARACTERISTICS OF COMPUTER FORENSICS

1. **Identification and Collection (Acquisition)** – Involves identifying and collecting digital evidence from electronic devices.
2. **Preservation** – Involves creating working copies (forensic images) and isolating or securing digital devices from unauthorized access.
3. **Overview** – Involves choosing the right method and approach for analysis.
4. **Analysis** – Involves the examination of data to extract relevant information in order to collect facts.
5. **Reporting** – Involves the documentation and presentation of evidence to legal bodies.

NEED FOR DIGITAL FORENSIC

1. It helps the companies **gather important information** on their computer systems or networks potentially being compromised.
2. To **produce evidence in the court**, which can lead to the punishment of the culprit.
3. Efficiently **tracks down cyber criminals** from anywhere in the world.
4. Helps to **protect** the organization's money and valuable time.

DIGITAL INVESTIGATION RULES AND REGULATIONS

1. **Preservation of Evidence** – No possible evidence is damaged, destroyed, or compromised by the forensic procedures used to investigate the computer.
2. **Prevention of Contamination of Evidence** – No possible computer malware is introduced to the computer being investigated during the analysis process.
3. **Accountability of Evidence** – A continuing chain of custody is established and maintained.
4. **Ethics of Investigation** – Details of the client-attorney relationship are not disclosed if obtained during a forensic process in order to maintain professional ethics and legality.
5. Normal operations are affected for a limited period of time, if at all.

ROLES & USES OF DIGITAL FORENSIC

Criminal Prosecutors	Civil Litigations	Insurance Companies	Large Corporations	Law Enforcement	Any Individual
Child Pornography cases	Fraud	False Accident Reports	Embezzlement		Sexual harassment
Homicides	Copyright	Workman's Compensation Cases	Insider Trading		Age discrimination
Embezzlement	Divorce				Wrongful termination from job
Financial Fraud	Breach of Contract				Background checks

KEY STEPS IN A FORENSIC INVESTIGATION

1. **Initial Response** – The investigation is initiated the moment the computer crime is suspected.
2. **Photographing The Crime Scene** – The immediate response is to collect preliminary evidence.
3. **Court Warrant** – A court warrant for seizure (if required) is obtained.
4. First responder procedures are performed.
5. **Seizure** – Evidence is seized at the crime scene. After seizure, the evidence is numbered and safely secured.
6. The evidence is securely transported to the forensic laboratory.
7. **Creating Forensic Disk Images** – Two bit-stream copies of the evidence are created. The original disk must not be tampered with as it might change the time stamps.

KEY STEPS IN A FORENSIC INVESTIGATION (CONTINUES…)

8. **Generate Hash Value (s)** – An MD5/SHA-256 checksum is generated on the images.
9. **Prepare a Chain of Custody** – Any change to this chain calls into questioning the admissibility of the evidence.
10. **Prepare a Forensic Report** – It describes the forensic method (s) and recovery tools used.

TASKS THAT ARE PERFORMED BY FORENSIC INVESTIGATOR

1. **Data Recovery** – Recovers data of investigative value from computers involved in crimes.
2. **Secure The Evidence** – Ensures that the evidence is not damaged in any way.
3. **Creating Forensic Image (s)** – Creates an image of the original evidence without tampering with it to maintain the original evidence's integrity.
4. **Prepares a Forensic Report**
5. **Provides Expertise Testimony** – Addresses the issue in a court of law and attempts to win the case by testifying in court.

LOCARD'S EXCHANGE PRINCIPLE

- States "**in the physical world, when perpetrators enter or leave a crime scene, they will leave something behind and take something with them**".
- Examples include **DNA**, **latent prints**, **hair**, and **fibers**.
- In digital forensics, **registry keys** and **log files** can serve as footprints left by the perpetrator (s).

ORGANIZATIONS INVOLVED IN DIGITAL FORENSICS

1. **Scientific Working Group on Digital Evidence (SWGDE) of 1998**

- Is made up of "federal government agency, state or local law enforcement agency involved in the digital and multi-media forensic profession".
- The federal Bureau of Investigation has supported formation of:

i. Scientific Working Groups (SWGs)

ii. Technical Working Groups (TWGs)

iii. Scientific Working Group for DNA Analysis Methods (SWGDAM)

iv. Scientific Working Group for Firearms and Toolmarks (SWGGUN)

ORGANIZATIONS INVOLVED IN DIGITAL FORENSICS (CONTINUES…)

2. **American Academy of Forensic Sciences (AAFS)**

- Members of the Academy work for:

i. National Institute of Standards and Technology (NIST)

ii. National Academy of Sciences (NAS).

iii. Scientific Working Group on Digital Evidence (SWGDE).

- It has approximately six thousand members and is divided into "eleven sections spanning the forensic enterprise."

ORGANIZATIONS INVOLVED IN DIGITAL FORENSICS (CONTINUES…)

3. **American Society of Crime Laboratory Directors/Laboratory Accreditation Board**

- Is the "oldest and most well known crime/forensic laboratory accrediting body in the world."

ORGANIZATIONS INVOLVED IN DIGITAL FORENSICS (CONTINUES…)

4. **National Institute of Standards and Technology (NIST) of 1901**

- It was the first federal physical science research laboratory.
- It is heavily involved in digital forensics. Some of the programs and projects include:

i. National Initiative Cyber Security Education (NICE)

ii. National Software References Library

iii. Computer Forensic Tool Testing

ORGANIZATIONS INVOLVED IN DIGITAL FORENSICS (CONTINUES…)

5. **American Society for Testing and Materials (ASTM) of 1898**

- Comprises about 30,000 members broken into 141 committees.
- The Forensics Sciences committee (E30), is further divided into several subcommittees.

i. The Digital and Multimedia Evidence (E30.12)

COMPUTER/DIGITAL FORENSICS TOOLS

- The goal of Computer forensics is to perform crime investigations by using evidence from digital data to find who was the responsible for that particular crime.

1. **Digital Forensics Framework (DFF)** – Open Source

- Has a command line and graphical interface.
- Is used to investigate hard drives, volatile memory, create reports about user and system activities.
- Can also be used for digital chain of custody, to access the remote or local devices, forensics of Windows or Linux OS, recovery of hidden deleted files, quick search for files' meta data, and various other things.

COMPUTER/DIGITAL FORENSICS TOOLS (CONTINUES…)

i. **Bulk Extractor (DFF)**

- Scans the disk images, file or directory of files to extract useful information.

ii. **Bulk Extractor Viewer (BEViewer)**

- Is a User Interface for browsing features that have been extracted from the Bulk Extractor forensic tool.

COMPUTER/DIGITAL FORENSICS TOOLS (CONTINUES…)

2. **Volatility Framework**

- Is an open source memory forensics tool written in Python.
- Is a command-line tool that is supported on Windows, MacOS and Linux.

3. **FTK Imager**

- Is a data preview and imaging tool used to acquire data (evidence) in a forensically sound manner by creating copies of data without making changes to the original evidence.

COMPUTER/DIGITAL FORENSICS TOOLS (CONTINUES…)

4. **Autopsy**

- Is a digital forensics platform and graphical interface to The Sleuth Kit® and other digital forensics tools.
- Is used by law enforcement, military, and corporate examiners to investigate what happened on a computer.
- You can even use it to recover photos from your camera's memory card.

COMPUTER/DIGITAL FORENSICS TOOLS (CONTINUES…)

5. **Encase**

- Is a digital forensic and investigation software that helps investigators collect, preserve, and analyze electronic evidence.
- Was developed by Guidance Software and OpenText in 1998.

FOOTPRINTS

- A **digital footprint** is any information about a particular person that exists on the internet as a result of their online activity.

1. **Active Digital Footprint –** Is data that you intentionally share online.

i. Posting on social media

ii. Completing online forms, such as for account registration or subscribing to a newsletter

iii. Writing reviews or leaving feedback for services

iv. Sending emails

FOOTPRINTS (CONTINUES…)

2. **Passive Digital Footprint**

- Refers to the trail of data a person leaves behind online without intentionally sharing it.

i. IP address

ii. Browsing history

iii. location data collected by websites

iv. Information gathered through cookies

QUESTIONS

1. Define Digital Forensics? What is difference in role of a CFI and FI?
2. Explain uses of Digital Forensics
3. What is Locard's Exchange Principle
4. What is the role of Organization in CFI?
5. List major organization governing standards of Digital forensics.
6. What will happen if the organizations like NIST, SWGDE, AAFS and ASTM does not exist?
7. What is Role of the Digital Forensic Examiner in the Judicial System?

ANSWERS

Q1.1

- **CFI (Certified Forensic Investigator)** – is a forensic investigator who has earned a recognized certification, demonstrating advanced expertise in forensic practices, particularly in areas like fraud and financial crimes.
- **FI (Forensic Investigator)** – is a general term for any investigator working in the forensic field, but they may not have formal certification or the same specialized knowledge as a CFI.

ANSWERS (CONTINUES…)

Q4

- The role of an Organization in CFI is to **provide certification, set professional standards**, and **offer ongoing training and resources to support forensic investigators** in maintaining their expertise and ethical practices.

ANSWERS (CONTINUES…)

Q6

- ❑ Lack of Standardization
- ❑ Inconsistent Quality and Accuracy
- ❑ Limited Professional Development
- ❑ Weak Legal Framework

ANSWERS (CONTINUES…)

Q7

- ❑ Collection and Preservation of Digital Evidence
- ❑ Analysis of Digital Evidence
- ❑ Expert Testimony
- ❑ Ensuring Chain of Custody

FILE CARVING AND DATA RECOVERY

LECTURE 3

COMPUTER DATA

- **Computer data** is information processed or stored by a computer.
- This information may be in the form of;

i. text documents,

ii. images,

iii. audio clips,

iv. software programs etc.

- Computer data may be processed by the computer's **CPU** and is stored in **files** and **folders** on the computer's hard disk.

DATA RECOVERY

- **Data recovery** is the process of retrieving lost, deleted, corrupted, or inaccessible data from a storage device, often using specialized software and techniques to scan the device, identify recoverable data, and restore it to a usable format.
- **Uses** of data recovery include;

i. Recover important lost files

ii. Locate illegal data

iii. Prosecute criminals based on discovered data

SOFTWARE DATA RECOVERY

- **Software data recovery** refers to the process of using specialized computer programs to retrieve lost, deleted, or damaged data from a storage device like a hard drive, USB stick, or memory card, by scanning the device for remnants of the data and attempting to restore it to a usable state.
- Tools may include;

i. Windows File Recovery

ii. EaseUS Data Recovery Wizard

iii. Stellar Data Recovery

BAD TRACK OF A HARD DISK

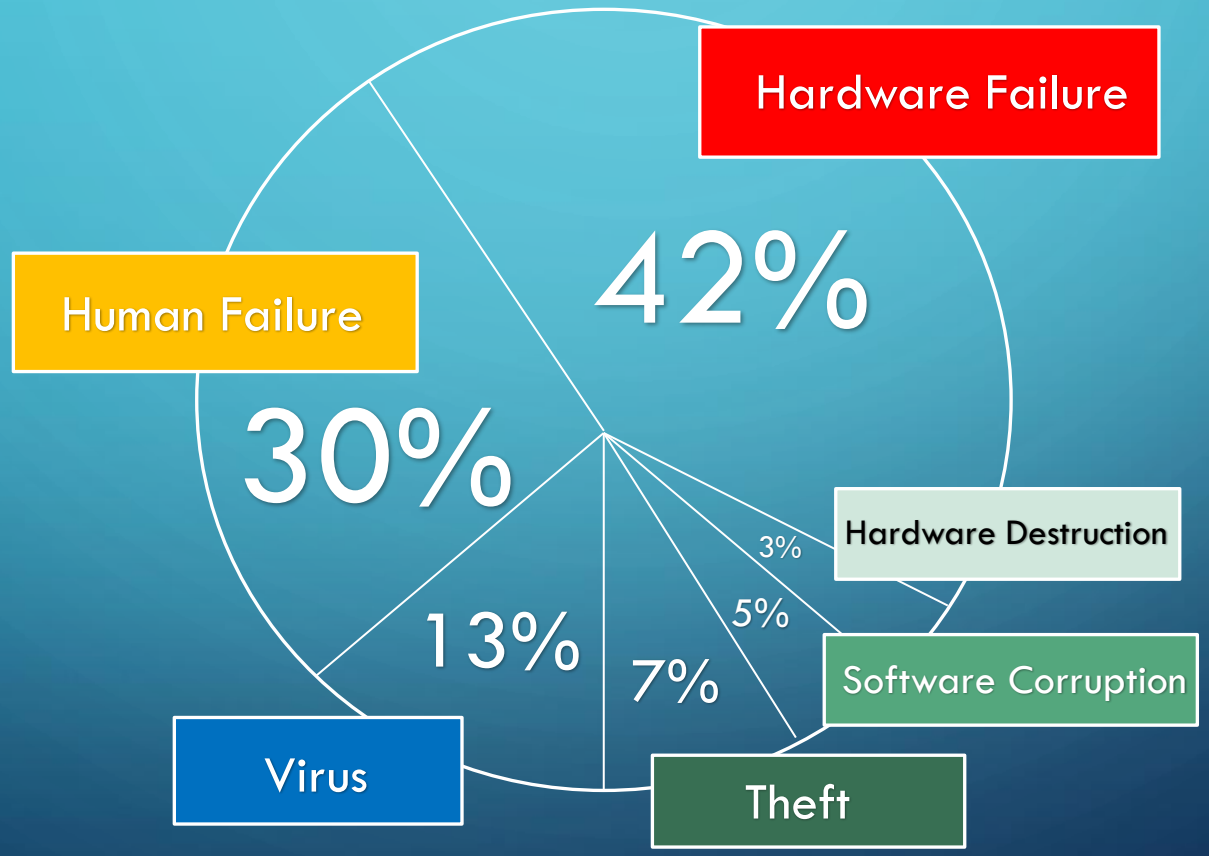
1. **Logical** – is mainly caused by incorrect operation, and it can be restored by software.
2. **Physical** – is caused by physical damage, which is real damage, we can restore it by changing the partition or sector.

RECOVERY METHODS

1. Recycle bin
2. Using data recovery software
3. Restoring data from backups stored on external drives or in the cloud

DATA LOSS

- Reasons…



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HARD DISK

- Is a magnetic storage medium for a computer.
- When hard disk is firstly made in the factory, it is usually “blank”.
- Only after partitioning tracks and sectors, we can save data on hard disk.



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DATA STORAGE REGION OF A HARD DRIVE

	MBR (63)	DBR (32)	FAT1	FAT2	DIR(32)	DATA
INTIALS	Master Boot Directory	Dos Boot Directory	File Allocation Table 1	File Allocation Table 2	Directory	Data
DESCRIPTION	Stores partition table and bootloader	Contains boot code, BIOS Parameter Block (BPB), and FAT structure details	Maps file allocation and tracks used/unused clusters	Backup copy of FAT1 for redundancy	Stores metadata (filenames, attributes, timestamps) for files and directories	Holds actual file contents, organized into clusters
SIZE	512 bytes	512 bytes	Varies (depends on partition size)	Same as FAT1	Fixed size in FAT12/16, dynamic in FAT32	Remaining space on the disk

FILE DELETION

- When deleting a file, the system only makes a deletion mark on this file's directory entry, marking clusters it covers in FAT as "empty"; clusters in DATA remains original file's contents.
- There is a **Recycle Bin** in Windows, which automatically establishes a folder "RECYCLED" (under root directory of each disk partition) with hiding attribute to save temporarily deleted files.

FILE DELETION (CONTINUES…)

1. **Fast Deletion**

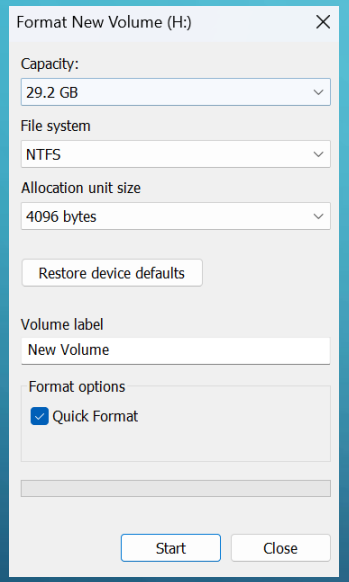
- Is just to put them into Recycle Bin.
- In this situation, the data can be recovered.

2. **Complete Deletion**

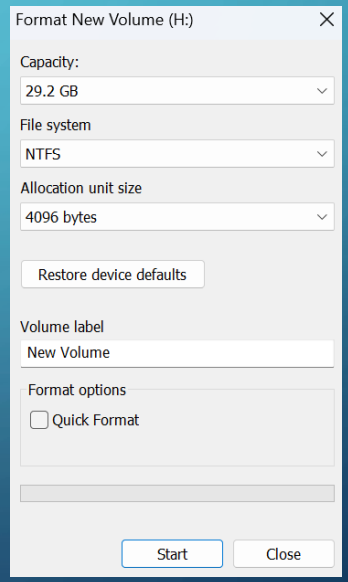
- Refers to the process of permanently removing a file from a storage device, where the data is completely overwritten, making it virtually impossible to recover using standard data recovery tools.

HIGH LEVEL FORMATTING

FAST HIGH LEVEL FORMATTING



COMPLETE HIGH LEVEL FORMATTING



FAT VS. NTFS

	File Allocation Table (FAT)	New Technology File System (NTFS)
Developed By	Microsoft (1977)	Microsoft (1993)
Maximum File Size	4GB (FAT32), 2GB (FAT16)	16TB (with default cluster size) or 256TB
Maximum Volume Size	2TB (FAT32), 4GB (FAT16)	8PB (Petabytes) or 256TB
Security Features	None (Basic file attributes)	Supports file encryption and access control (ACLs)
Performance	Slower for large files	Faster due to file compression and indexing
Used in	USB drives, memory cards, older systems	Modern Windows systems, SSDs, enterprise storage

DELETION TECHNIQUES

1. **Multiple Overwrites**
2. **Extreme Physical Destruction**

i. Chainsaws

ii. Drop in a volcano

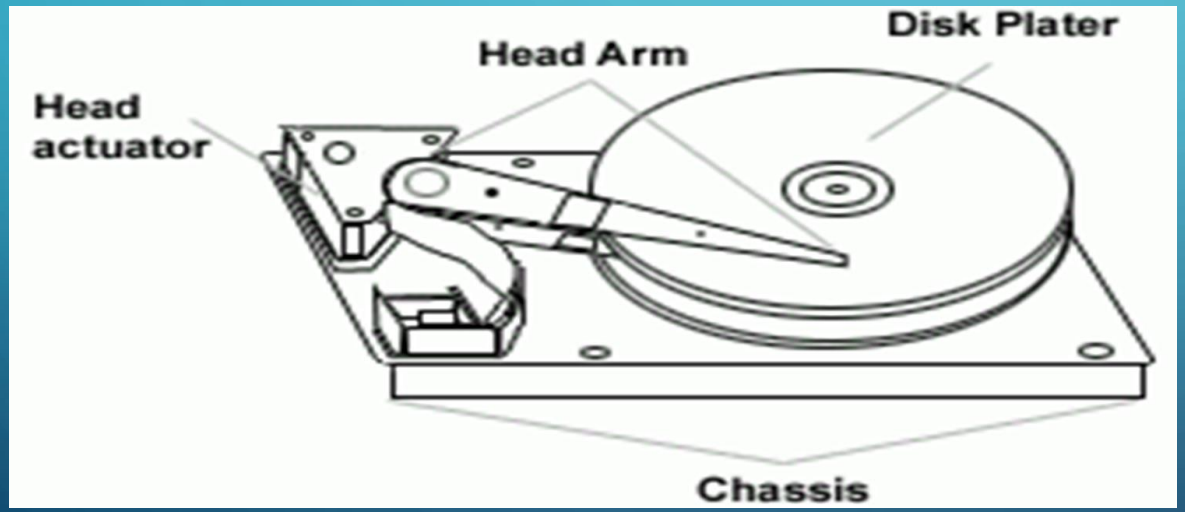
iii. Sledge hammers

HARD DISK DRIVE (HDD)

LECTURE 4

MAIN COMPONENTS FOR HARD DISK DRIVE

- HDD were invented in 1954 by IBM.
- It's main components include –



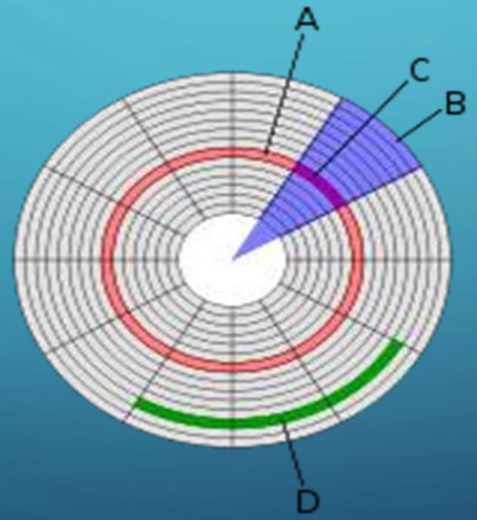
MAIN COMPONENTS FOR HARD DISK DRIVE (CONTINUES…)

1. **Disk Platter** – is made up of a magnetic material, in the flat disk part of the drive and it stores data.
2. **Stepper Motor** – is used for controlling read/write head position.
3. **Spindle Motor** – controls the platter.

READ AND WRITE HEAD

- The heads read and write the information to the drive platter.
- The head arm is used for read and write operations.
- Below is a disk structure –

A. Track
B. Sector
C. Cylinder
D. Storage Capacity



PARTITION FOR HDD

1. **Primary Partition** – used to boot the operating system.
2. **Extended Partition** – can be subdivided into multiple logical partitions.
3. **Logical Partition** –
4. **Active Partition** –

TYPES OF INTERFACEING IN HDD

1. **Integrated Device Electronics (IDE)** – also known as Parallel Advanced Technology Attachment (PATA).
2. **Serial Advanced Technology Attachment (SATA)** – It is the latest high-speed type of hard drive connectors.
3. Small Computer System Interface (SCSI) –

HDD CAPACITY AND SIZE INFORMATION

Term	Equal To
Bit	0 or 1
Kilo Bit (kb)	1024 bits
Byte	8 bits
Kilobyte (KB)	1024 bytes
Megabyte (MB)	1024 kilobytes
Gigabyte (GB)	1024 megabytes
Terabyte (TB)	1024 gigabytes
Petabyte (PB)	1024 terabytes
Exabyte (EB)	1024 petabytes
Zettabyte (ZB)	1024 exabytes
Yottabyte (YB)	1024 zettabyte

MULTI-MEDIA FORENSICS

LECTURE 5.1

CLASSIFICATION OF MULTIMEDIA

1. **Text** - ASCII/Unicode, HTML, PDF
2. **Audio** – Sound, music, speech
3. **Still Image** - photo, scanned image
4. **Video (Moving Images)** – Movie
5. **Graphics** – Computer produced image
6. **Animation** – A sequence of graphics images
7. **Discrete Media (DM, Static)** – text, image, graphics
8. **Continuous Media (CM, Dynamic)** – audio, video, animation
9. **Captured vs Synthesized media**
10. **Standalone vs Networked media**

MULTIMEDIA

- **Multimedia** is a combination of text, sound, graphics, video, and animation delivers by computer or any other electronic media.



USE AREA OF MULTIMEDIA

1. **Business** – application for multimedia include presentation, Training, advertisement, video, conferencing and etc.
2. **Education** – Multimedia is a better form of teaching in schools and colleges.
3. **Home**
4. **Public**

ELEMENTS OF MULTIMEDIA

1. **Text** – can have different font and size to suit the presentation.
2. **Image** – It is the most important and interesting component of multimedia.
3. **Sound** – It is categorized by frequency ,web length, amplitude, speech etc. **Mono** uses a single audio channel and applied in public address systems e.g., podcasts, while **stereo** uses two channels (left and right) and applied in music, movies, video games etc.
4. **Animation** – animation makes static presentations come alive.
5. **Video** – It is the technology of capturing, recording, processing, transmitting, and reconstructing moving pictures.

MULTI-MEDIA FORENSICS (AUDIO & VIDEO)

LECTURE 5.2

INTRODUCTION

- Unlike other forms of forensic evidence, **audio** and **video** recordings can provide a real-time, eyewitness account at a crime so investigators can watch or hear what transpired.
- **Authentication of recordings –**

i. Lighting conditions can be examined to estimate the time of day or environmental conditions at the same time of the recording.

ii. Unnatural waveform present in the audio or video signal may indicate that an edit has been made.

TYPES OF VIDEO AND AUDIO

1. Digital

- Half of all Closed-Circuit Television (CCTV) evidence seized by the police today is digital.
- Digital video recorders come in two general types, **embedded stand-alone** and **PC-based**, both types generally record the audio and video to Hard Drives, Secure Digital (SD) cards.
- Digital video and audio **recording devices** include CDs/DVDs and USB.

TYPES OF VIDEO AND AUDIO (CONTINUES…)

2. Analog

- Every playback of the tape will degrade the recorded images.

IMPORTANT EVIDENCE TO BE RECORDED

i. Make and model of the recording device.
ii. How many cameras are connected and recording.
iii. Whether it is recording in time-lapse mode.
iv. The current time/date, and the time/date on the recorder's display
v. Cameras' positions.

WHO CONDUCTS THE ANALYSIS

- **American Society of Crime Laboratory Directors / Laboratory Accreditation Board** differentiates the audio and/or video analysis into four parts –

1. Forensic audio analysis
2. Forensic video analysis
3. Image analysis
4. Computer forensics

HOW THE ANALYSIS IS PERFORMED

i. First step is to simply listen to or view the recorded footage.

ii. Locate the area of interest to be enhanced or examined using specialized devices and software.

iii. Before processing a working copy of the evidence may be created.

VIDEO ENHANCEMENT TECHNIQUE

1. **Sharpening** – Makes edges of images in the recording become more clear and distinct.
2. **Stabilization** – Reduces the amount of movement in the video, producing the smoothest possible playback.
3. **Masking** – Covers the face or areas of the video that may protect a witness, victim or law enforcement officer.
4. **Demultiplexing** – Allows for isolation of each camera.

AUDIO ENHANCEMENT TECHNIQUES

1. **Frequency Equalization** – Highly precise equalizers can be used to boost (using **amplification**) or cut specific bands of frequencies and high noisy sound can be analyzed by a **spectrum analyzer**.
2. **Compression** – Faint sounds in the recording can be boosted by compressing.

MALWARE FORENSICS

LECTURE 6

MALWARE

- The word **MALWARE** comes from the combination of **Malicious** and **Software.**
- In Spanish, '**MAL**' is a prefix that means, '**BAD**' , making the term '**bad ware**'.
- Malware is program that is specifically designed to gain access or damage a computer without the knowledge of the owner.
- It's purpose include **stealing, encrypting, deleting, altering** and **monitoring** user data.

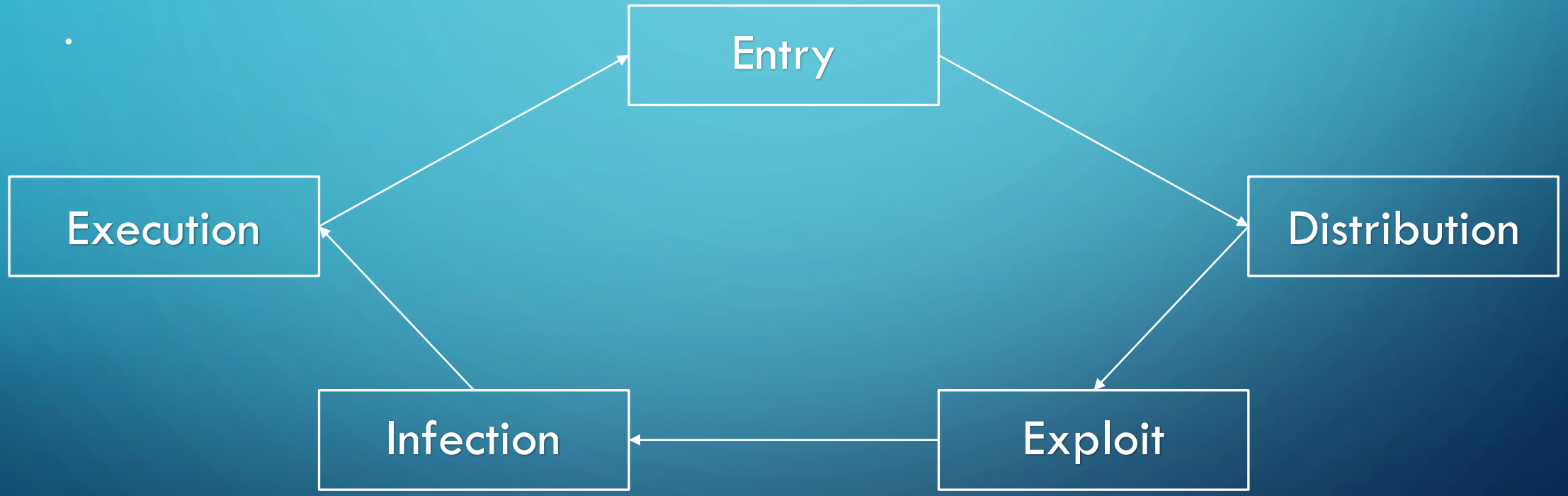
HISTORY OF MALWARE

- People first started writing malware in the 1970s and early 1980s. They were written as experiments or pranks for fun.
- '**Creeper Virus**' created in 1971 was an experiment which did not cause damage but did foretell the future of malware with its quick spread through systems.
- '**Reaper**' was created to hunt and destroy Creeper, one of the first examples of an '**anti-virus**' program.
- Term '**malware**' was coined in 1990 and its use increased in the early 2000s.

DISTRIBUTION OR INFECTION METHODS OF A MALWARE

a) Delivered physically to a system through a storage media device such as an USB drive.
b) Drive-by-Downloads via the Internet.
c) Spam E-mails and Phishing Attacks.
d) Bundled with other software.
e) Accessing hacked or compromised webpages.

THE 5 STAGES OF MALWARE ATTACK



TYPES OF MALWARE

1. **Virus**

- Is a computer program that infects a host computer and then spreads when executed.

i. **Boot Sector** – infects the boot sector of disk storage (MBR).

ii. **Macro** – exploits the macro language of a program like Microsoft Word or Excel.

iii. **Logic/Time Bomb** – malicious code embedded in a program that executes when certain conditions are met.

TYPES OF MALWARE (CONTINUES…)

2. Worms

- are very similar to a virus, but it can send copies of itself to other nodes on a computer network without having to be carried by an infected host file.

i. **Email** – spreads itself through an email and can hide itself in messages as a link or an attachment that will redirect the user to an infected website.

ii. **Instant Messaging** – a link that will redirect the user to an infected website will be shared though an IM like WhatsApp.

TYPES OF MALWARE (CONTINUES…)

3. **Trojan Horse**

- is a program that appears harmless, but is, in fact, malicious.

i. **Remote Access Trojans (RATs)** – allow attackers to take control of the victim's computer remotely.

ii. **DDoS Trojans** – turn infected machines into bots to flood target servers.

TYPES OF MALWARE (CONTINUES…)

iii. Rootkit

- It is a computer program designed to provide continued privileged access to a computer while actively hiding its presence.
- Originally, a rootkit was a collection of tools that enabled administrator-level access to a computer or network.

TYPES OF MALWARE (CONTINUES…)

iv. Fake Antivirus Trojans (Scareware)

- is a type of malware designed to trick victims into purchasing and downloading useless and potentially dangerous software.

TYPES OF MALWARE (CONTINUES…)

v. Ransomware

- Is a type of malicious software (malware) that encrypts a victim's files or locks their system, making data inaccessible until a ransom is paid, usually in cryptocurrency (e.g., Bitcoin).
- Originated in Russia, 2005-2006.
- Types of Ransomware include;

a) **Crypto-ransomware (2013)** – Encrypts files (e.g., WannaCry, Locky, CryptoLocker).

b) **Locker Ransomware** – Locks access to the computer (e.g., WinLocker).

c) **Double Extortion** – Steals data before encryption and threatens to leak it if ransom isn't paid (e.g., Maze, REvil).

TYPES OF MALWARE (CONTINUES…)

How It Works

a) **Infection** – Often spread via phishing emails, malicious links, or infected downloads.

TYPES OF MALWARE (CONTINUES…)

vi. Spyware

- is a program that get installed without the user's permission to monitor the user's activities on the internet and transmit the information to the third party.
- Types include;

a) Tracking cookies
b) System monitors
c) Commercial cell phone spywares
d) Keyloggers

TYPES OF MALWARE (CONTINUES…)

vii. Adware

- is a software where advertising banners are displayed while any program is running.

PROTECTING AGAINST MALWARE

1. **Anti-malware/virus software**

- Are computer programs intended to identify and eliminate computer viruses.
- For example, **Microsoft Security Essentials** (for Windows XP, Vista, and Windows 7) and **Windows Defender** (for Windows 8, 10 and 11) provides real-time protection.
- The **Windows Malicious Software Removal Tool** removes malicious software from the system.

PROTECTING AGAINST MALWARE (CONTINUES…)

- Example of anti-virus softwares include;

i. BitDefender
ii. Norton AntiVirus (NAV)
iii. McAfee
iv. Sophos
v. Kasperski

PROTECTING AGAINST MALWARE (CONTINUES…)

Types of anti-virus

a) Specific Scanning (signature detection)

- The application scans files to look for known viruses matching definitions in a "**virus dictionary**".
- After recognizing the malicious software the antivirus software can take one of the following actions;
- **Repair, quarantine** or **delete** the file completely.
- A signature-based is completely **blind to new attacks** that are yet to be recorded.

PROTECTING AGAINST MALWARE (CONTINUES…)

b) Generic Scanning (suspicious behavior approach)

- is used when new viruses appear, in this method the software does not look for a specific signature but instead monitors the behavior of all applications.
- If anything questionable is found by the software the application is **quarantined** and a warning is broadcasted to the user about what the program may be trying to do.
- If the software is found to be a virus the user can send it to a virus vendor so that they release the antivirus software to stop its spread.

CLOUD FORENSICS

UNIT 7

TYPES OF CLOUDS

- **Cloud computing** is a technology that uses the internet and central remote servers to maintain data and applications.
- Types of cloud include;

1. **Public** – The cloud is open to the wider public.
2. **Private** – The cloud is implemented within the private premises of an institution.

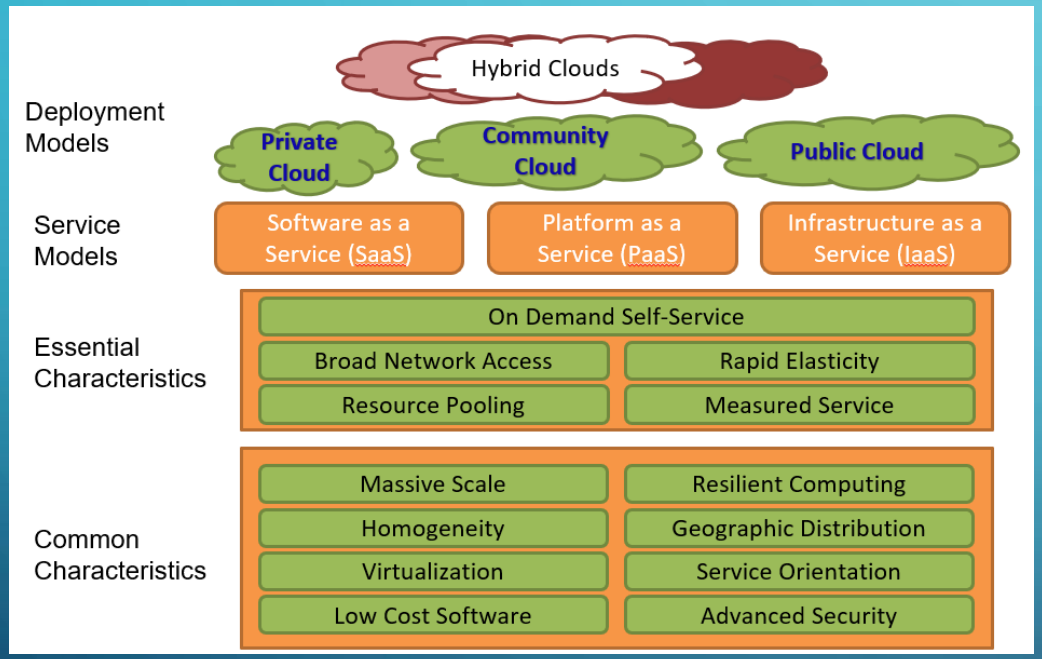
TYPES OF CLOUDS (CONTINUES…)

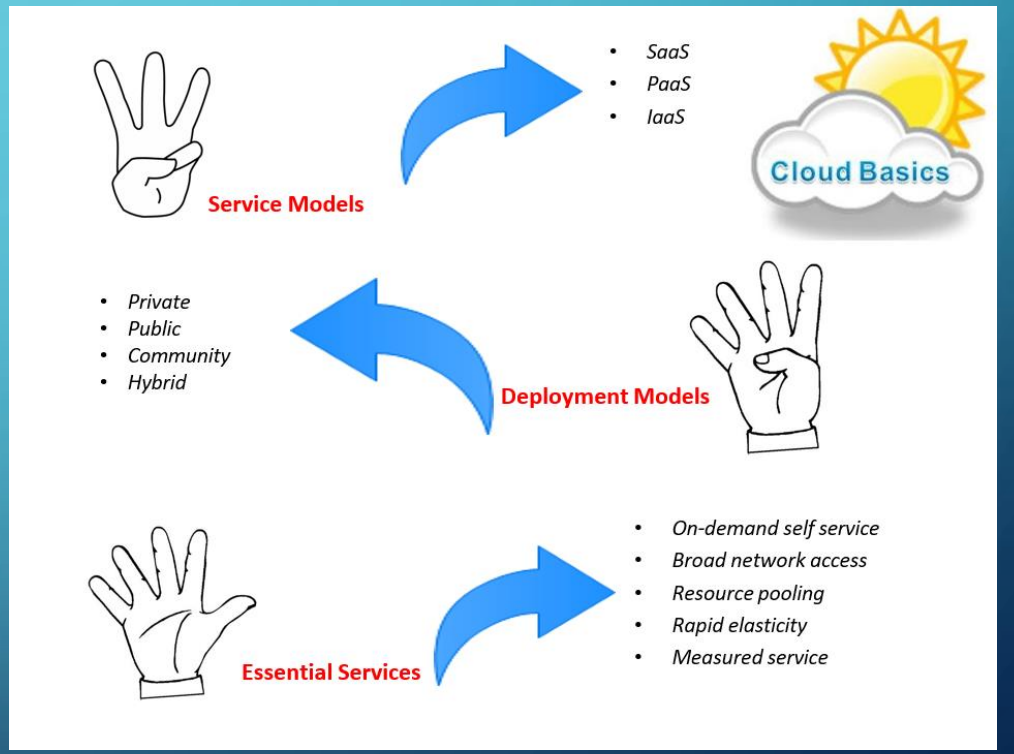
3. **Hybrid/Heterogeneous** – A combination of both Public and Private clouds.
4. **Community** – The cloud is characterized by a multi-administrative domain involving different deployment models (public, private, and hybrid), and it is specifically designed to address the needs of a specific industry.

CLOUD SERVICE MODELS

Service Model	Who Uses It	Available Services	Why Use It
SaaS	Agency Staff Members	Software applications such as email	Complete business tasks that are typically performed locally on a computer
PaaS	Developers and Application Managers	Service for creating, testing, managing, and hosting software applications	Establish a common and consistent platform for application development
IaaS	IT System Managers	Virtual machines, storage and backup services	Build a customized computing environment

CLOUD COMPUTING FRAMEWORK





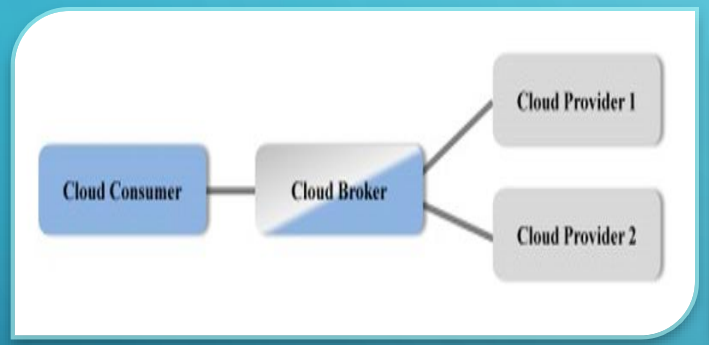
FORENSICS

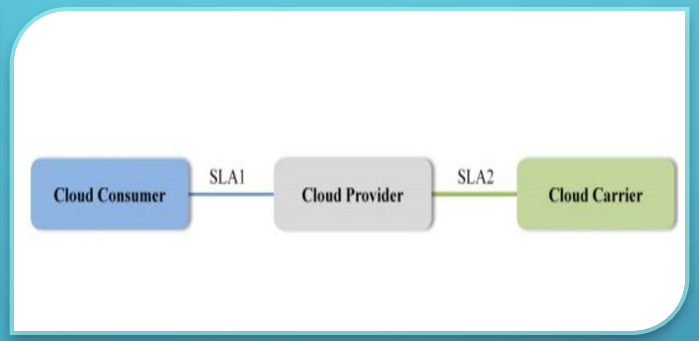
- **Digital forensics** involves the collection, preservation, and analysis of digital evidence.
- **Cloud forensics** is the application of digital forensics science in cloud computing environments.
- **Technically,** it consists of a **hybrid forensic approach** (e.g., **remote, virtual, network, live**) towards the generation of digital evidence.
- **Organizationally,** it involves **interactions** among cloud actors (i.e., **cloud provider, consumer, broker, carrier, auditor**) for the purpose of facilitating both internal and external investigations.
- **Legally,** it often implies **multi-jurisdictional** and **multi- tenant** situations.

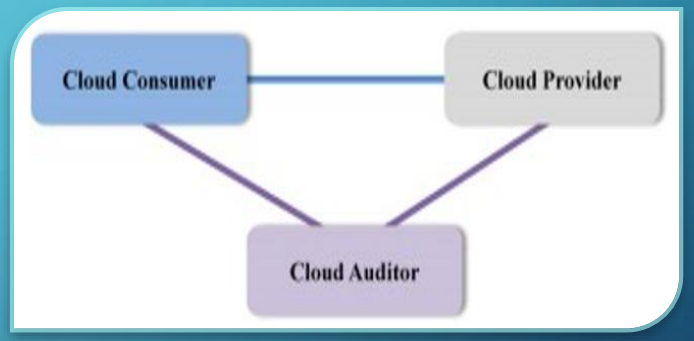
ACTORS IN CLOUD COMPUTING

1. **Consumer** – A person or organization that maintains a business relationship with, and **uses services** from the Cloud Providers.
2. **Provider** – A person, organization, or entity responsible for **making a service available** to interested parties.
3. **Auditor** – A party that can **conduct independent assessment of cloud services**.
4. **Broker** – An entity **negotiates relationships** between Cloud Providers and Cloud Consumers.
5. **Carrier** – An **intermediary** that **provides connectivity** and **transport** of cloud services from Cloud Providers to Cloud Consumers.

SCENARIOS IN CLOUD







SCENARIO 1

Cloud consumer interacts with the cloud broker instead of contacting a cloud provider directly.

SCENARIO 2

Cloud carriers provide the connectivity and transport of cloud services from cloud providers to cloud consumers.

SCENARIO 3

Cloud auditor conducts independent assessments for the operation and security of the cloud service.

ACTORS IN CLOUD COMPUTING (CONTINUES…)

- **Service Level Agreements (SLAs)** cover the quality of service, security, remedies for performance failures.
- **Cloud Providers** offer services like SaaS, PaaS and IaaS.
- **Audits** are performed to verify conformance to standards.
- Integration of cloud services can be complex for consumers. Hence cloud broker, is needed.

ACTORS IN CLOUD COMPUTING (CONTINUES…)

- In general, a cloud broker can provide services in three categories:

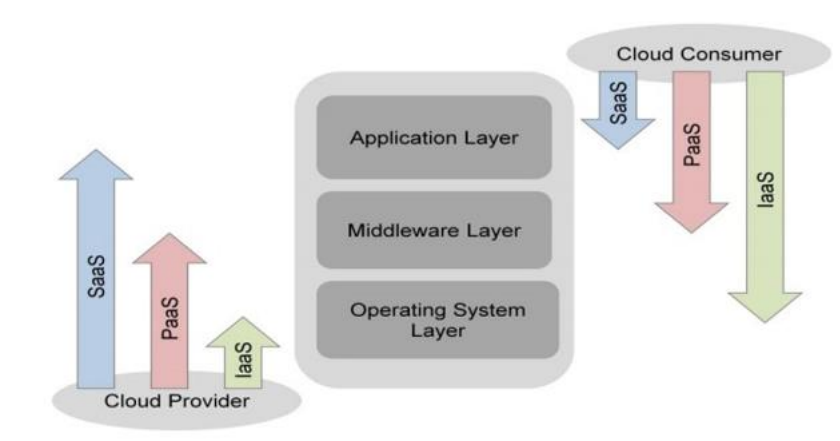
i. **Service Intermediation –** broker (a third-party service or system) stands between the consumer and the actual service provider to enhance or improve the service being delivered.

ii. **Service Aggregation –** Broker combines and integrates multiple services into one or more new services (mash up).

iii. **Service Arbitrage –** It is similar to service aggregation with the flexibility to choose services from multiple agencies.

ACTORS IN CLOUD COMPUTING (CONTINUES…)

- The distribution of cloud services is normally provided by **network** and **telecommunication carriers** or a transport agent.
- **Transport agent** refers to a business organization that provides physical transport of storage media such as high-capacity hard drives.



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MEMORY FORENSICS

LECTURE 8

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INTRODUCTION

- While a system is on, **Random Access Memory (RAM)** contains critical information about the current state of the system.
- By capturing an entire copy of RAM and analyzing it on a separate computer, it is possible to reconstruct the state of the original system, including the applications the user was running and the files or network connections that existed at the time.
- Processes, files and programs are run in memory (**volatile information**), and a **snapshot** (memory dump) can be captured by a forensic investigator.

WHAT IS MEMORY FORENSICS?

- **Memory forensics** refers to finding and extracting forensic artifacts from a computer's physical memory.
- Below are some differences between memory forensics and disk forensics.

Feature	Memory Forensics	Hard Disk Forensics
Image	Snapshot (memory dump)	Disk image and mostly used for data revery
State	Live response	Post mortem
Volatility	Data is volatile and can be lost when the machine is powered off	Non-volatile data

WHY MEMORY FORENSICS?

1. **Hiding Data** – Attackers may hide data in the memory by means of memory injection and it only resides in memory not hard disk.

DATA ACQUISITION METHODS IN MEMORY FORENSICS

1. **Hardware-based acquisition**

- Suspends the processor and uses direct memory access to obtain a copy of memory when the **PCI** card is plugged in the computer system and a copy of the memory is transferred to an external storage device.
- The card is installed into a PCI bus slot before an incident occur.
- It is more **reliable** and difficult for an attacker to corrupt, however, they are **costly**.

DATA ACQUISITION METHODS IN MEMORY FORENSICS (CONTINUES…)

2. Software-based acquisition

- Is often carried out using a **trusted toolkit** by forensic examiners.
- This method can also collect volatile memory content using tools built in the operating system.
- Drawbacks of software-based acquisition are that it **may alter contents of memory**, and potentially overwrite data relevant to an investigation.
- However, they are **cost-effective**.

FORMATS OF DATA ACQUIRED IN MEMORY FORENSICS

1. **RAW** – extracted from a live environment.
2. **Crash Dump** – information gathered by the operating system.
3. **Hibernation File** – a saved snapshot that your operating system can return to after hibernating.
4. **VMWare Snapshot** – this is a snapshot of a virtual machine, which saves its state as it was at the exact moment that the snapshot was generated.
5. **Page File** – this is a file that stores similar information that is stored in your system RAM.

MEMORY FORENSIC TOOLS

1. **Volatility Suite –** is an open source suite of programs for analyzing RAM, and has support for Windows, Linux and MacOS.
2. **Rekall –** This is an end-to-end solution for incident responders and investigators, and features both acquisition and analysis tools.
3. **Helix ISO –** This is a bootable live CD as well as a standalone application that makes it very easy for you to capture a memory dump or memory image of a system.
4. **Belkasoft RAM Capturer –** This is another forensic tool that allows for the volatile section of system memory to be captured to a file.

THESE NOTES ARE 85% DONE, YOU ARE ADVISED ON CHECK THE HANDOUT OF CLOUD FORENSICS AND THE NEED FOR MEMORY FORENSICS.